A Two-Atom Counterexample to the Nonlinear-Phase Exponential Basis Question

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Abstract

Gabardo, Lai, and Oussa ask whether every finite Borel measure μ can admit an orthogonal basis of generalized exponentials $E(\Lambda, \varphi) = \{e^{2\pi i \lambda \cdot \varphi(x)} : \lambda \in \Lambda\}$ for some Borel phase φ. We give a literal counterexample: an unequal two-atom probability measure. The obstruction is that generalized exponentials have modulus one at every atom, so two such vectors can be orthogonal in a two-atom weighted L^2 space only when the two atom weights are equal.

1 Source Question

The source paper is:

> Jean-Pierre Gabardo, Chun-Kit Lai, and Vignon Oussa, *On exponential bases and frames with non-linear phase functions and some applications*, arXiv:2007.03736.

Section 8, "Open problems", asks:

> Given any finite Borel measure μ on $\mathbb{R}^d$. Due to the flexibility of Borel measurable functions φ, is it true that every $L^2(\mu)$ can admit some $E(\Lambda, \varphi)$ as an orthogonal basis? Can we find a Borel measure that does not admit any orthogonal basis of non-linear phase?

The rendered source page is included in Figure 1.

2 Definitions

For a countable $\Lambda \subset \mathbb{R}^d$ and a Borel measurable phase φ, the generalized exponential system is

$$E(\Lambda, \varphi) = \{x \mapsto e^{2\pi i \lambda \cdot \varphi(x)} : \lambda \in \Lambda\}.$$

In an atomic $L^2(\mu)$ space, these are simply vectors with all coordinates of modulus one.

3 Proof Intuition

The phase has enormous freedom, but it cannot change the weights of atoms. For a measure with two atoms of unequal masses, $L^2(\mu)$ is a two-dimensional weighted Hilbert space. An orthogonal basis would require two nonzero orthogonal generalized exponentials. Since both vectors have

(1) Given any finite Borel measure μ on $\mathbb{R}^d$. Due to the flexibility of Borel measurable functions φ, is it true that every $L^2(\mu)$ can admit some $E(\Lambda,\varphi)$ as an orthogonal basis? Can we find a Borel measure that does not admit any orthogonal basis of non-linear phase?

(2) The middle-third Cantor measures example given in Section 4 admits orthogonal basis of exponentials with a non-linear phase function φ. They are not differentiable everywhere and is drastically different from our familiar phase functions $\varphi(x) = x$. A natural question here is that can we find φ, defined in an open set containing the support, of better regularity (e.g., φ is C^∞) so that $E(\Lambda,\varphi)$ forms an orthogonal basis? For other open, connected sets, do we have an exponential orthogonal basis with a non-linear phase?

(3) Theorem 5.1 provides a characterization on $\mathbb{R}^1$ that a continuous function can be an orthogonal basis for $L^2[0,1]$ with integer frequencies, provided that φ preserves measure zero sets. It looks like that there may exist a continuous function, which will be highly irregular, such that $E(\mathbb{Z},\varphi)$ forms a basis or a frame for $L^2[0,1]$. Will there be any such function? Or can we remove the preserving measure-zero set assumption in Theorem 5.1?

(4) Theorem 6.1 provides a large class of functions that $E(\mathbb{Z}^d,\varphi)$ can form an orthogonal basis for $L^2[0,1]^d$. Are there any other C^1-functions φ not of the form

$$x \mapsto M\left(x_1 + l_1\left(x_2,\cdots x_d\right), x_2 + l_2\left(x_3,\cdots,x_d\right),\cdots, x_{d-1} + l_{d-1}\left(x_d\right), x_d\right)$$

for some C^1-functions $l_1, l_2, \cdots, l_{d-1}$ where M is a matrix in integer entries satisfying $|\det M| = 1$ for which $E(\mathbb{Z}^d,\varphi)$ is an orthonormal basis for $L^2[0,1]^d$?

References

Figure 1: Source open-problem page from arXiv:2007.03736, Section 8.

coordinate moduli equal to one, their inner product is a sum of two complex numbers whose moduli are exactly the two atom masses. Two complex numbers can sum to zero only if their moduli are equal. Unequal atom weights therefore prevent even a pair of orthogonal generalized exponentials.

Lemma 1. *Let $\mu = a\delta_{x_0} + b\delta_{x_1}$ with $a, b > 0$ and $a \neq b$. For every Borel measurable $\varphi : \{x_0, x_1\} \to \mathbb{R}^d$ and every countable $\Lambda \subset \mathbb{R}^d$, the system $E(\Lambda,\varphi)$ contains no two orthogonal nonzero vectors in $L^2(\mu)$.*

Proof. Take two frequencies $\lambda, \lambda' \in \Lambda$. Their inner product is

$$\left\langle e^{2\pi i\lambda\cdot\varphi}, e^{2\pi i\lambda'\cdot\varphi}\right\rangle_{L^2(\mu)} = ae^{2\pi i(\lambda-\lambda')\cdot\varphi(x_0)} + be^{2\pi i(\lambda-\lambda')\cdot\varphi(x_1)}.$$

The two summands have moduli a and b. If the inner product were zero, then

$$ae^{2\pi i(\lambda-\lambda')\cdot\varphi(x_0)} = -be^{2\pi i(\lambda-\lambda')\cdot\varphi(x_1)}.$$

Taking absolute values gives $a = b$, contradicting the hypothesis. Thus no two distinct generalized exponentials are orthogonal. ∎

Theorem 1. *There exists a finite Borel measure μ on $\mathbb{R}$ such that no Borel phase φ and no countable frequency set Λ make $E(\Lambda,\varphi)$ an orthogonal basis for $L^2(\mu)$.*

Proof. Let

$$\mu = \frac{1}{3}\delta_0 + \frac{2}{3}\delta_1.$$

Then $L^2(\mu)$ is two-dimensional. Any orthogonal basis of this Hilbert space must contain two nonzero orthogonal vectors. By the lemma, no generalized exponential system $E(\Lambda, \varphi)$ contains such a pair. Hence no such system is an orthogonal basis for $L^2(\mu)$. $\square$

4 Verification Notes

The proof is independent of regularity of the phase; it applies to every Borel phase and every countable frequency set. It also covers linear phases, so it is stronger than merely excluding nonlinear phases.

Equivalently, using the source's Theorem 3.2, an essential-injective phase would push the two unequal atom weights to a two-atom measure with the same unequal weights. A finite atomic spectral measure with two atoms must have equal weights, since an orthogonal pair of modulus-one vectors in the two-dimensional weighted space exists only in that case.

5 Novelty Check

Before promotion, the run searched the local registry for arXiv:2007.03736 and the title/key phrases, finding no existing packet for this question. Web queries used the exact open-question wording and close variants involving "orthogonal basis of non-linear phase", "Borel measure", "unequal atoms", "spectral measure", and "exponentials". They found the source paper but no later explicit answer. This packet should therefore be reviewed as a likely new, literal counterexample rather than as a literature retrieval.

6 Scope and Limitations

This answers the first open problem as literally stated for arbitrary finite Borel measures. It does not address the remaining open problems in Section 8: regular nonlinear phases for Cantor-type supports or open connected sets, the one-dimensional continuous irregular phase problem for Lebesgue measure, or classification of higher-dimensional C^1 phases on cubes.

7 Human Review Recommendation

Review should focus on whether the intended problem silently assumes nonatomic or continuous measures. If not, the unequal two-atom measure gives a complete negative answer.

References

[1] J.-P. Gabardo, C.-K. Lai, and V. Oussa, *On exponential bases and frames with non-linear phase functions and some applications*, arXiv:2007.03736, 2020.